5 hours on examples

Problem 1 (03JPN1)

P is on a circle centered at F with diameter AB so $\angle APB = 90°$. Now, we see that,

$$\angle A = \angle BPA + \angle PAD = 90° - \angle ADB + \angle BAP = 3x - 90°$$

This means,

$$BP = AB \cdot \sin(x) = b$$

We scale things such that the diameter of (BFP) is one unit.

We let $x = \angle FPB = \angle ACF$ and $y = \angle BFP$. Clearly, by law of sines,

$$CP = AF = BF = \sin(x)\,;$$

$$BP = \sin(y)\,;$$

$$FP = \sin(x+y)$$

Now doing law of sines on AFC,

$$1 + \frac{\sin(x+y)}{\sin(x)} = \frac{CF}{AF} = \frac{\sin(\angle FAC)}{\sin(\angle ACF)} = \frac{\sin(\angle BEC - \angle ABE)}{\sin x} = \frac{\sin\big((180-2x)-(180-x-y)\big)}{\sin x}$$

$$\Leftrightarrow 1 + \frac{\sin(x+y)}{\sin(x)} = \frac{\sin(y-x)}{\sin(x)}$$

$$\Leftrightarrow \sin(x) = \sin(y-x) - \sin(x+y)$$

$$\Leftrightarrow 1 = -2\cos(y)$$

Clearly, $y = 120°$ is the only possible solution as desired.

Problem 3 (19EGMO3)

We let the intersection of the $\angle BAD$ bisector with BC be P_1 and that of $\angle BXC$ be P_2.

$$\frac{BP_1}{BC} = \frac{BP_1}{BD} \cdot \frac{BD}{BC} = \frac{AB}{AB+AD} \cdot \frac{a - CD}{a}$$

$$= \frac{c}{c + \frac{b}{a} \cdot c} \cdot \frac{a - \frac{b^2}{a}}{a} = \frac{1}{1 + \frac{b}{a}} \cdot \frac{a^2 - b^2}{a^2} = \frac{a}{a+b} \cdot \frac{a^2-b^2}{a^2} = \frac{a-b}{a}$$

Equivalently, $\frac{BP_1}{P_1C} = \frac{a-b}{b}$. We now work on $\frac{BP_2}{P_2C} = \frac{BX}{XC}$. Let BA intersect (AIX) again at Y. Note that, looking at $\overline{CAA}$ and $\overline{BYA}$, there is a spiral similarity at X sending BY to CA. Therefore,

$$\frac{BX}{XC} = \frac{BY}{CA} = \frac{BY}{b}$$

We show that $BY = a - b$ to finish. Indeed, since,

$$\angle IYA = \angle IXA = \angle IAC$$

We know that $AI = IY$. Therefore, if we let F be the base of the altitude from I to AB, $AY = 2AE = 2(s-a) = b + c - a$. Thus, $BY = AB - AY = a - b$.

5 clubs; 7 hours

Problem 4 (09IMO4)

We know $\angle B = \angle C = 90 - \frac{\angle A}{2}$. Therefore, $\angle BEC = 45 - \frac{\angle A}{4} + \angle A = 45 + \frac{3\angle A}{4}$. If $\angle IEK = 45$, then we know $\angle KEC = \frac{3\angle A}{4}$. Therefore,

$$\frac{IK}{KC} = \frac{IE}{EC} \cdot \frac{\sin 45}{\sin \frac{3\angle A}{4}} = \frac{\sin ECI}{\sin CIE} \cdot \frac{\sin 45}{\sin \frac{3\angle A}{4}} = \frac{\sin \frac{\angle C}{2}}{\sin BIC} \cdot \frac{\sin 45}{\sin \frac{3\angle A}{4}} = \frac{\sin \frac{\angle C}{2}}{\cos \frac{\angle A}{2}} \cdot \frac{\sin 45}{\sin \frac{3\angle A}{4}}$$

$$= \frac{\sin \frac{\angle C}{2}}{\cos \frac{\angle A}{2}} \cdot \frac{\sin 45}{\sin \frac{3\angle A}{4}} = \frac{\sin\left(45 - \frac{\angle A}{4}\right)}{\cos \frac{\angle A}{2}} \cdot \frac{\sin 45}{\sin \frac{3\angle A}{4}}$$

Also, since $DI \perp DC$, $\frac{IK}{KC} = \frac{DI}{DC} = \tan\frac{C}{2} = \tan\left(45 - \frac{\angle A}{4}\right)$. Therefore,

$$\frac{\sin\left(45 - \frac{\angle A}{4}\right)}{\cos \frac{\angle A}{2}} \cdot \frac{\sin 45}{\sin \frac{3\angle A}{4}} = \tan\left(45 - \frac{\angle A}{4}\right)$$

$$\Leftrightarrow \cos\left(45 - \frac{\angle A}{4}\right) \cdot \sin(45) = \cos\left(\frac{\angle A}{2}\right) \cdot \sin\left(\frac{3\angle A}{4}\right)$$

$$\Leftrightarrow \cos\left(45 - \frac{\angle A}{4}\right) \cdot \sin(45) = \cos\left(\frac{\angle A}{2}\right) \cdot \sin\left(\frac{3\angle A}{4}\right)$$

$$\Leftrightarrow \frac{\sin\left(90 - \frac{\angle A}{4}\right) + \sin\left(\frac{\angle A}{4}\right)}{2} = \frac{\sin\left(\frac{5\angle A}{4}\right) + \sin\left(\frac{\angle A}{4}\right)}{2}$$

$$\Leftrightarrow \sin\left(90 - \frac{\angle A}{4}\right) = \sin\left(\frac{5\angle A}{4}\right)$$

On $0 < \angle A < 180$, the solutions are $\angle A = 60, 90$.

Problem 5 (19SLG2)

We find the inradii of ω_B, ω_C. Note that triangles BFD, BCA are similar. Therefore,

$$r_{BFD} = r_{BCA} \cdot \frac{BF}{BC} = r_{BCA} \cdot \cos B$$

Where R is the circumradius of (ABC).

$$= 4R \sin\left(\frac{A}{2}\right) \sin\left(\frac{B}{2}\right) \sin\left(\frac{C}{2}\right) \cdot \cos B$$

Analogously, $r_{CDE} = 4R \sin\left(\frac{A}{2}\right) \sin\left(\frac{B}{2}\right) \sin\left(\frac{C}{2}\right) \cdot \cos C$. Note that, by law of sines,

$$PM = QN$$

$$\Leftrightarrow 2r_{BFD} \sin FMP = 2r_{CDE} \sin ENQ$$

$$\Leftrightarrow \cos B \sin FMP = \cos C \sin ENQ$$

$$\Leftrightarrow \frac{\sin FMP}{\sin ENQ} = \frac{\cos C}{\cos B}$$

Let N' be where the incircle of ω_C touches BC. Note that,

$$\frac{\sin FMP}{\sin ENQ} = \frac{\sin DMN}{\sin DNM} = \frac{DN}{DM} = \frac{DN'}{DM}$$

Since triangles DBF and DEC are similar, quadrilaterals $DBFM$ and $DECN'$ are similar. Therefore,

$$\frac{DN'}{DM} = \frac{CE}{BF} = \frac{\cos C}{\cos B}$$

As desired.

11 clubs; 8 hours

<u>Problem 7 (13JMO5)</u>

We let $\angle YXB = x$; $\angle BXA = y$. Therefore,

$$\frac{BY}{XP} = \frac{2R \sin x}{\frac{2R \sin(90 - x - y)}{\cos y}} = \frac{\sin x \cos y}{\cos(x + y)}$$

Since $\angle YXC = 90 - \angle AZX = 90 - \angle APX = y$,

$$\frac{CY}{XQ} = \frac{2R \cos y}{\frac{2R \sin(90 - x - y)}{\sin y}} = \frac{\sin y \cos x}{\cos(x + y)}$$

As a result,

$$\frac{BY}{XP} + \frac{CY}{XQ} = \frac{\sin(x + y)}{\cos(x + y)}$$

Indeed, this is $\tan(x + y) = \frac{AY}{AX}$.

Problem 8 (01IMO5)

Simply,

$$AB + BP = AQ + QB$$

$$\Leftrightarrow 2R\sin C + \frac{2R\sin C\sin 30}{\sin(B+30)} = \frac{2R\sin C\sin\left(\frac{B}{2}\right)}{\sin\left(C+\frac{B}{2}\right)} + \frac{2R\sin C\sin(60)}{\sin\left(C+\frac{B}{2}\right)}$$

$$\Leftrightarrow 1 + \frac{\sin 30}{\sin(B+30)} = \frac{\sin\left(\frac{B}{2}\right) + \sin(60)}{\sin\left(C+\frac{B}{2}\right)}$$

Note that $B + C = 180 - A = 120$.

$$\Leftrightarrow 1 + \frac{\sin 30}{\sin(B+30)} = \frac{\sin\left(\frac{B}{2}\right) + \sin(60)}{\sin\left(120-\frac{B}{2}\right)} \Leftrightarrow 1 + \frac{\sin 30}{\sin(B+30)} = \frac{\sin\left(\frac{B}{2}\right) + \sin(60)}{\sin\left(60+\frac{B}{2}\right)}$$

$$\Leftrightarrow 1 + \frac{\sin 30}{\sin(B+30)} = \frac{2\sin\left(\frac{B}{4}+30\right)\cos\left(\frac{B}{4}-30\right)}{2\sin\left(\frac{B}{4}+30\right)\cos\left(\frac{B}{4}+30\right)} \Leftrightarrow 1 + \frac{\sin 30}{\sin(B+30)} = \frac{\cos\left(\frac{B}{4}-30\right)}{\cos\left(\frac{B}{4}+30\right)}$$

$$\Leftrightarrow \frac{\sin 30}{\sin(B+30)} = \frac{\cos\left(\frac{B}{4}-30\right) - \cos\left(\frac{B}{4}+30\right)}{\cos\left(\frac{B}{4}+30\right)} \Leftrightarrow \frac{\sin 30}{\sin(B+30)} = \frac{2\sin\left(\frac{B}{4}\right)\sin(30)}{\cos\left(\frac{B}{4}+30\right)}$$

$$\Leftrightarrow \cos\left(\frac{B}{4}+30\right) = 2\sin(B+30)\sin\left(\frac{B}{4}\right)$$

$$\Leftrightarrow \cos\left(\frac{B}{4}+30\right) = \cos\left(-\frac{3B}{4}-30\right) - \cos\left(\frac{5B}{4}+30\right)$$

$$\Leftrightarrow \cos\left(\frac{B}{4}+30\right) + \cos\left(\frac{5B}{4}+30\right) = \cos\left(\frac{3B}{4}+30\right)$$

$$\Leftrightarrow 2\cos\left(\frac{3B}{4}+30\right)\cos\left(\frac{B}{2}\right) = \cos\left(\frac{3B}{4}+30\right) \Leftrightarrow \cos\left(\frac{3B}{4}+30\right)\left(2\cos\left(\frac{B}{2}\right)-1\right) = 0$$

We find that the only solution on $B < 120$ is $B = 80$. The angles are $A = 60; B = 80; C = 40$.

$$\Leftrightarrow \cos(50) = \cos(-90) - \cos(130)$$

$$\Leftrightarrow \cos(50) = 2\sin(110)\sin(20)$$

$$= \sin(\cdot \frac{\sin(-10) - \cos(50)}{\sin(30)}$$

$$1+\frac{\sin(30)}{\sin(110)}=\frac{\cos(-10)}{\cos(50)}$$

16 clubs; 10 hours

Problem 10 (19TSTST5)

(Is this the triglen solution? I read each solution in the overleaf and didn't see prominently triglen)

Let P be where ray HK intersects (ABC) again and Q be where ray KH intersects (ABC). We show that,

$$DP\perp BC \Leftrightarrow DP \parallel AH \Leftrightarrow \frac{PK}{KH}=\frac{DK}{KA}$$

$$\Leftrightarrow \frac{PK}{DK}=\frac{KH}{KA}\Leftrightarrow \frac{AK}{KQ}=\frac{KH}{KA}\Leftrightarrow AK^2=KH\cdot KQ \Leftrightarrow KE^2=KH\cdot KQ$$

Claim 1: KE is tangent to (BEH)

Proof: $\angle FEK=90-A=\angle HBE.$ □

Similarly, KF is tangent to (HFC). Note that K,H hence both lie on the radical axis of $(BEH),(HFC)$. Therefore, the second intersection of these two circles lies on KH. However, by Miquel's theorem on AEF, their intersection lies on (ABC) meaning it is Q.

$(BEHQ)$ is cyclic. As a result, $KE^2=KH\cdot KQ$ as desired.

Problem 11 (17SLG1)

We equivalently show that the line through P perpendicular to BC passes through E. Let G be the base of the altitude from P to BC. Let $E_1=DE\cap GP; E_2=AE\cap GP$. We show that $PE_1=PE_2$ to finish.

$$AB=BC=CD=1; \angle B=\angle D=2y; \angle C=\angle A=2x$$

By isosceles triangles, $\angle CAB=\angle BCA=90-y; \angle DBC=\angle CDB=90-x$. Angle chasing, we'd get $\angle BPA=180-x-y; \angle ABP=2y+x-90$. Therefore,

$$AP=\frac{\sin(2y+x-90)}{\sin(180-x-y)}$$

Looking at APE_2,

$$PE_2=\frac{AP\cdot\sin(PAE_2)}{\sin(AE_2P)}=\frac{AP\cdot\sin(2x+y-90)}{\sin(270-2x-2y)}=\frac{\sin(2y+x-90)\cdot\sin(2x+y-90)}{\sin(180-x-y)\cdot\sin(270-2x-2y)}$$

This is symmetric in x,y, so when we calculate PE_1, we will get the same result.

22 clubs; 12 hours

Problem 13 (12SLG3)

Claim: I the incenter of ABC lies on the radical axis of $(AI_1C), (BI_2C)$.

Proof: Note that $AI_1 \cap BI_2 = I$. Therefore, it suffices to prove,

$$II_1 \cdot IA = II_2 \cdot IB$$

WLOG, let the inradius of ABC be $r = 1$. Note that, since AEF and ABC are similar triangles, the inradius r_1 of AEF would be $r_1 = r \cdot \frac{AF}{AC} = \cos(A)$. Therefore,

$$AI_1 = \frac{r_1}{\sin\left(\frac{A}{2}\right)} = \frac{\cos(A)}{\sin\left(\frac{A}{2}\right)}$$

Given that $AI = \frac{r}{\sin\left(\frac{A}{2}\right)} = \frac{1}{\sin\left(\frac{A}{2}\right)}$, we have $II_1 = \frac{1-\mathrm{c}\quad(A)}{\sin\left(\frac{A}{2}\right)}$. Therefore,

$$II_1 \cdot IA = \frac{1-\cos(A)}{\sin\left(\frac{A}{2}\right)} \cdot \frac{1}{\sin\left(\frac{A}{2}\right)} = \frac{1-\cos(A)}{\sin^2\left(\frac{A}{2}\right)} = \frac{1-\cos(A)}{\sin^2\left(\frac{A}{2}\right)}$$

$$= \frac{1-\cos^2\left(\frac{A}{2}\right)+\sin^2\left(\frac{A}{2}\right)}{\sin^2\left(\frac{A}{2}\right)} = \frac{\cos^2\left(\frac{A}{2}\right)+\sin^2\left(\frac{A}{2}\right)-\cos^2\left(\frac{A}{2}\right)+\sin^2\left(\frac{A}{2}\right)}{\sin^2\left(\frac{A}{2}\right)} = 2$$

Analogously,

$$II_2 \cdot IB = \frac{1-\cos(B)}{\sin^2\left(\frac{B}{2}\right)} = 2 \quad \square$$

Therefore,

$$O_1O_2 \parallel I_1I_2 \Leftrightarrow CI \perp I_1I_2$$

We may angle chase to finish noting that (AI_1I_2B) is cyclic using Theorem 2.9.

<u>Problem 14 (23MEX3)</u>

$$AB \cdot CD = 4PM \cdot PK \Leftrightarrow MB \cdot KC = PM \cdot PK$$

$$\Leftrightarrow \frac{MB}{PK} = \frac{MP}{CK} \Leftarrow \Delta BMP \sim \Delta PKC$$

We show this similarity to finish. Note that,

$$\frac{BP}{PC} \cdot \sin(\angle BPN) \cdot \sin(\angle NPC) = \frac{BN}{NC} \Leftrightarrow \frac{BP}{PC} \cdot \sin(\angle BPN) \cdot \sin(\angle NPC) = 1$$

$$\Leftrightarrow \frac{BP}{PC} = \frac{\sin(\angle NPC)}{\sin(\angle BPN)} \Leftrightarrow \frac{BP}{PC} = \frac{\sin(\angle PDA)}{\sin(\angle PAD)} \Leftrightarrow \frac{BP}{PC} = \frac{PA}{PD}$$

Now note that,

$$\angle BPC = \angle PAD + \angle PDA \Leftrightarrow \angle BPC + \angle APD = 180$$

$$\Leftrightarrow \angle BPA + \angle DPC = 180 \Leftrightarrow \angle BPA = \angle CDP + \angle DCP$$

Claim: $\angle BPM = \angle KCP$.

Proof: Note that,

$$\frac{BP}{PC} = \frac{PA}{PD} \Leftrightarrow \frac{BP}{PA} = \frac{CP}{PD} \Leftrightarrow \frac{\sin(\angle MPB)}{\sin(\angle APM)} = \frac{\sin(\angle CDP)}{\sin(\angle PCD)}$$

Note that the value $\frac{\sin(\angle CDP)}{\sin(\angle PCD)}$ is fixed whereas $\frac{\sin(\angle M'PA)}{\sin(\angle BP\ \ ')}$ is strictly increasing as a point M' "slides" from A to B since the sum $\angle M'PA + \angle M'PB = \angle BPA = \angle CDP + \angle DCP$ is fixed. Therefore,

$$\Leftrightarrow \angle MPB = \angle DCP; \angle APM = \angle PCD \Longrightarrow \angle BPM = \angle KCP$$

Symmetrically, we have $\angle CPK = \angle ABP; \angle DPK = \angle PAB \Longrightarrow \angle CPK = \angle MBP$. Using these two, we have triangles MBP and KPC are similar as desired.

28 clubs; 15 hours

<u>Problem 16 (24USATST2)</u>

Claim: $BP = BQ$.

Note that since $\angle PAB = 90 - \angle BAI = 90 - \frac{A}{2}$, we have,

$$BP = AB \cdot \sin(\angle PAB) = c \cdot \cos\left(\frac{A}{2}\right)$$

On the other hand, we may calculate using $BQ \perp AC$,

$$\angle QBA = 90 - A$$

Using tangency,

$$\angle BAQ = \angle IBQ = \frac{B}{2} - (90 - A) = A + \frac{B}{2} - 90$$

Therefore, $\angle AQB = 180 - \frac{B}{2}$. We may use law of sines,

$$\frac{AB}{\sin(\angle AQB)} = \frac{BQ}{\sin(\angle BAQ)}$$

$$\Leftrightarrow BQ = \frac{c \cdot \sin\left(A + \frac{B}{2} - 90\right)}{\sin\left(180 - \frac{B}{2}\right)} = \frac{c \cdot \sin\left(\frac{A}{2} - \frac{C}{2}\right)}{\sin\left(\frac{B}{2}\right)}$$

However, we also know $\sin\left(\frac{B}{2}\right) = \frac{IE}{BI} = \frac{ID}{BI} = \frac{\sin(\angle IBD)}{\sin(\angle BDI)} = \frac{\sin\left(\frac{A-C}{2}\right)}{\sin\left(90\ \ \frac{A}{2}\right)}$.

$$\Leftrightarrow BQ = \frac{c \cdot \sin\left(\frac{A}{2} - \frac{C}{2}\right)}{\frac{\sin\left(\frac{A-C}{2}\right)}{\sin\left(90 - \frac{A}{2}\right)}} = c \cdot \sin\left(90 - \frac{A}{2}\right) = BP \quad \square$$

With the claim proven, note that,

$$\angle IBP = 180 - \angle AIB = 90 - \frac{C}{2} = \frac{A}{2} + \frac{B}{2}$$

Subtracting $\angle IBQ$ from this,

$$\angle PBQ = \frac{A}{2} + \frac{B}{2} - A - \frac{B}{2} + 90 = 90 - \frac{A}{2} = \frac{B}{2} + \frac{C}{2}$$

$$\Leftrightarrow \angle BPQ = \frac{180 - \angle PBQ}{2} = 90 - \frac{B}{4} - \frac{C}{4}$$

$$\Leftrightarrow \frac{B}{4} + \frac{C}{4}$$

We may angle chase to get $\angle PAX = \frac{3A}{4} + \angle PAB = \frac{3A}{4} = 90 + \angle \frac{A}{4}$ meaning the remaining angle $\angle AXP$ of the triangle has measure 45 as desired.

Problem 19 (Russian Killer)

Lemma: This works if $AB + CD = BC + AD$.

Proof: The quadrilateral inscribes a circle by Pitot. Where the touch points on AB, BC, CD, DA are W, X, Y, Z, we have,

$$r = \frac{AW}{\cot\left(\frac{A}{2}\right)} + \cdots + \frac{DZ}{\cot\left(\frac{D}{2}\right)}$$

$$= \frac{AW + \cdots + DZ}{\cot\left(\frac{A}{2}\right) + \cdots + \cot\left(\frac{D}{2}\right)}$$

$$= \frac{\frac{1}{2}(AB + \cdots + DA)}{\cot\left(\frac{A}{2}\right) + \cdots + \cot\left(\frac{D}{2}\right)}$$

We have,

$$[ABCD] = \left(\frac{1}{2} \cdot r \cdot AB\right) + \cdots + \left(\frac{1}{2} \cdot r \cdot DA\right)$$

$$= \frac{\frac{1}{4}(AB + \cdots + DA)^2}{\cot\left(\frac{A}{2}\right) + \cdots + \cot\left(\frac{D}{2}\right)} \quad \square$$

We draw the angle bisectors of $\angle A, \dots, \angle D$ to form a quadrilateral $PQRS$ outside of $ABCD$. We slide A' along the segment of $PQRS$ which A lies on and define B', C', D' to be other points on the corresponding segments of $PQRS$ such that $AB \parallel A'B'$, … We slide A' until we get to a point such that $A'B' + C'D' = B'C' + D'A'$. This is possible since sliding A' in one direction, $A'D'$ and $B'C'$ increase while $A'B'$ and $D'C'$ decrease.

Let h be the distance from A' to AB. By congruent triangles, h is the distance from B' to BC and so on. Drawing the altitude from A', B' to AB, we get that,

$$A'B' = AB - h\tan\frac{A}{2} - h\tan\frac{B}{2}$$

$$C'D' = CD - h\tan\frac{C}{2} - h\tan\frac{D}{2}$$

$$B'C' = BC + h\tan\frac{B}{2} + h\tan\frac{C}{2}$$

$$D'A' = DA + h\tan\frac{D}{2} + h\tan\frac{A}{2}$$

Therefore, the perimeters of $ABCD$ and $A'B'C'D'$ are the same. Note that by Pitot's,

$$A'B' + C'D' = B'C' + D'A'$$

$$\Leftrightarrow h = \frac{AB + CD - BC - DA}{\tan\frac{A}{2} + \tan\frac{B}{2} + \tan\frac{C}{2} + \tan\frac{D}{2}}$$

Therefore, since,

$$[A'B'C'D'] - [ABCD]$$

$$= h \cdot (A'B' - BC) + h \cdot (B'C' - BC) + h \cdot (C'D' - AD) + h \cdot (D'A' - AD)$$

$$= h \cdot (AB + CD - BC - DA)$$

$$= \frac{(AB + CD - BC - DA)^2}{\tan\frac{A}{2} + \tan\frac{B}{2} + \tan\frac{C}{2} + \tan\frac{D}{2}}$$

We are done using the lemma.

43 clubs; 20 hours